



Research paper

Implicit rigid tube model predictive control for underwater manipulators with adaptive sliding mode strategy

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ABSTRACT

This paper presents an innovative control scheme combining implicit rigid tube model predictive control (IRTMPC) with adaptive sliding mode control (ASMC) for the trajectory tracking of underwater manipulators, addressing the challenges posed by model uncertainties and external disturbances in underwater environment. The IRTMPC approach avoids the complexity of explicit set algebra operations by utilizing implicit minimal robust invariant set representations, enabling efficient computation suitable for high-dimensional systems. The auxiliary tube control law in IRTMPC is designed by discrete ASMC, reinforcing the system's stability and precision. Meanwhile, the dynamic parameters of a newly designed underwater manipulator are identified using a least squares identification method based on the integral operator. Finally, the effectiveness of the proposed IRTMPC-ASMC scheme for controlling the underwater manipulator is validated through trajectory tracking control simulation and experiments.

1. Introduction

As the development and utilization of ocean resources continue to increase, underwater manipulators have been used in various complex underwater operations. Ruan et al. (2024) introduces a heavy-duty hydraulic manipulator capable of operating at full ocean depths. Mao et al. (2023) presents a lightweight underwater manipulator specifically designed for delicate structural repair tasks. In Zhang et al. (2021) the research team develops a new type of piezoelectric-actuated two-degree-of-freedom (2-DOF) joint for underwater manipulators, which is highly suitable for underwater tasks requiring lightweight design and fine manipulation. Ge et al. (2024) explores the trajectory tracking control of a two-joint underwater manipulator in ocean-wave environments. Simetti et al. (2021) achieves autonomous detection, tracking, landing, and in-situ measurement of seabed mineral nodules through the integration of control and perception frameworks in the underwater vehicle manipulators system.

The control and operation of underwater manipulators on complex underwater environment is a great challenge due to ocean currents, complex hydrodynamic effects, and the highly nonlinear and uncertain nature of the underwater manipulators. To address uncertainties and disturbances in underwater environments, (Han et al., 2021) introduced an adaptive sliding mode control method combined with an extended state observer. This method significantly enhances system robustness by real-time estimation and compensation of external

disturbances, ensuring precise tracking performance of the manipulator in complex underwater conditions. Zhou et al. (2022) proposed an adaptive control strategy combining backstepping and robust controllers. This method effectively handled the dynamic complexities of hydraulic systems and uncertainties in underwater environments, improving control precision. Fan et al. (2022) introduced a kinematics-based data-driven control scheme that achieves good control performance without requiring an accurate mathematical model, particularly suitable for dynamic underwater operations. Yu et al. (2021) and Han et al. (2022) developed adaptive control methods based on fuzzy logic. These methods achieve high-precision trajectory tracking control by approximating system uncertainties with fuzzy systems and adapting well to sudden changes in the underwater environment. To minimize disturbances caused by manipulators during underwater operations, Li and Li (2021) proposed a sparse optimization-based control method. This method minimizes the output torque changes of the manipulator while ensuring task accuracy, thereby improving system stability and energy efficiency.

However, due to high pressure and the need for sealing underwater, the position and torque constraints of underwater motors are often very stringent, meaning that input constraints or optimality need to be considered when controlling underwater manipulators. Model predictive control (MPC) can meet these requirements while ensuring control accuracy. To address constraints due to dynamic uncertainties,

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sensor measurement noise, and time-varying external disturbances, Dai et al. (2020) proposed a control scheme combining adaptive extended Kalman filtering (EKF) with fast model predictive control (FMPC). By combining Gaussian process regression (GPR) with MPC, Hewing et al. (2020), Xu et al. (2023), Janssen et al. (2020), Zhang et al. (2022) directly assess system model uncertainties through Gaussian process (GP) models and achieve MPC control based on the learned non-parametric models. Neural network-based MPC methods estimate system models and solve optimization problems through online learning (Kang et al., 2021; Chen et al., 2022).

The above methods that compensate MPC controllers through model learning or data-driven approaches require large datasets, and small-scale model errors still exist. Nubert et al. (2020), Zanon and Gros (2021) handled system uncertainties through constructed tube channels based on machine learning, achieving effective control of complex dynamic systems. Tube MPC has gained increasing attention for its robustness, and its advantage lies in the ability to simultaneously handle optimal control and error control separately. Dai et al. (2022) proposed a fast tube-MPC scheme based on sliding mode control, combining the fast response characteristics of sliding mode control with the robustness of tube-MPC. Kohler et al. (2021) introduced a computationally efficient robust model predictive control framework for handling uncertain nonlinear systems. By combining tube-MPC with event-triggered mechanisms, Yang et al. (2024) reduced computational burdens by eliminating unnecessary control updates, while tube-MPC handled system uncertainties, achieving efficient disturbance suppression and trajectory tracking. By constructing state tubes, Zheng et al. (2020) effectively dealt with external disturbances such as ocean currents and model uncertainties, achieving high-precision positioning control. Robust model predictive control for constrained linear systems under bounded disturbances introduced tube-MPC strategies to effectively handle system dynamics, model uncertainties, and external disturbances, achieving high-precision trajectory tracking of actuators (Mayne et al., 2005; Dai et al., 2021; Fleming et al., 2015; Hao et al., 2024).

However, tube-MPC requires explicit calculation and maintenance of rigid tubes and their terminal constraint sets, often involving complex set algebra operations, particularly in high-dimensional systems, significantly increasing computational complexity and potentially making tube-MPC infeasible. Raković (2023) innovatively proposed implicit rigid tube model predictive control (IRTMPC), which uses implicit set representations, avoiding explicit set algebra operations in optimization problems. This allows optimization problems to be directly solved by standard solvers such as LINPROG and QUADPROG, significantly improving computational efficiency and making tube-MPC more feasible in high-dimensional systems. This paper presents an innovative control scheme combining IRTMPC with adaptive sliding mode control (ASMC) for the trajectory tracking of underwater manipulators, leveraging the robustness of SMC and the constraint optimality of MPC to enhance the robustness of underwater manipulator control while ensuring control optimality.

Furthermore, parameter identification is crucial in robotics and control systems, being the key to improving dynamic model accuracy and control performance. Accurate parameter identification allows researchers to better understand system behavior characteristics, laying a solid foundation for designing efficient and stable controllers. Recent research, such as the enhanced integral method proposed by Li et al. (2023), considers the complexity of nonlinear robotic systems and optimizes for quantization errors, significantly improving parameter identification accuracy. Calanca et al. (2011) and Zhang et al. (2024) explored parameter identification techniques from the perspectives of multi-input multi-output closed-loop systems and hydraulic manipulators, enriching research in this field. Additionally, Zhou et al. (2024) and Sousa and Cortesao (2019) introduced nonlinear friction models and linear matrix inequality constraints, demonstrating parameter identification methods considering physical feasibility conditions. This

paper adopts a least-squares identification method based on integral operators to identify the dynamic parameters of an actual underwater manipulator, providing an accurate dynamic model for controller design. Finally, the following contributions were obtained:

- This paper identifies the parameters of a newly designed underwater manipulator and proposes an excitation function parameter optimization algorithm based on an improved genetic algorithm. The parameters are identified using a least squares identification method based on integral operators.
- This paper proposes combining a discrete-time implicit rigid tube robust MPC strategy and discrete adaptive SMC to control the underwater manipulator. The discrete-time implicit rigid tube robust MPC reduces the computational load of the minimal robust invariant set, and the tube in MPC is designed by discrete adaptive SMC.
- Simulation and field experiments are provided to verify the feasibility of the proposed scheme, and the state constraints of the underwater manipulator and the constrained input of the control law are satisfied.

The rest of this paper is organized as follows. Sections 2 and 3 introduce the symbols and trajectory tracking control problem of underwater manipulator. In Section 4, the unknown inertial parameters are estimated, and Section 5 details the implicit rigid tube model predictive control for underwater manipulators and its auxiliary tube ASMC law, analyzing its system convergence. Section 6 provides simulation and experimental validation of the proposed control scheme. Finally, Section 7 concludes the paper.

2. Notations

The notations used in this paper are described as follows: The sets of real numbers and non-negative integers are denoted by $\mathbb{R}$ and $\mathbb{N}$, respectively. $N_{[a:b]}$ represents the set of positive integers from a to b . $\sigma_{\max/\min}^S(A)$ represents the maximum/minimum singular value of the matrix A . The operation $a \oplus b$ denotes the Minkowski sum of a and b . $h(A, b) := \sup_a \{b^T a : a \in A\}$ denotes the support function for the vector b in the non-empty, closed convex subset A .

3. Problem formulation

This section presents a newly designed three-degree-of-freedom underwater manipulator with the design structure and the actual prototype shown in Fig. 1. The dynamic equations of the underwater manipulators can be expressed as follows:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + F(q, \dot{q}) + \tau_d = \tau \quad (1)$$

where $q \in \mathbb{R}^n$, $\dot{q}$ and $\ddot{q}$ represent the joint position vector, joint velocity vector, and joint acceleration vector of the underwater manipulator, respectively. n represents the degrees of freedom of the manipulator. In addition, the inertia matrix $M(q) \in \mathbb{R}^{n \times n}$, the centripetal and Coriolis matrix $C(q, \dot{q}) \in \mathbb{R}^{n \times n}$, the gravity vector $G(q) \in \mathbb{R}^n$, and the friction matrix $F(q, \dot{q}) \in \mathbb{R}^n$ respectively describe the system's inertial properties, non-inertial force effects, gravitational influence, and frictional resistance. $\tau \in \mathbb{R}^n$ represents the joint torque and $\tau_d \in \mathbb{R}^n$ represents the external disturbance torques acting on each joint, a portion of τ_d is the fluid dynamic torque. In this paper, the various links of the manipulators are simplified as cylinders, and Morrison's formula is utilized to establish the fluid resistance torque τ_r , and added mass torque τ_m experienced by each joint of the manipulators:

$$\tau_r = \frac{1}{2} \rho C_r D_i L_i \int_0^l v^n(x) \|v^n(x)\| dx \quad (2)$$

$$\tau_m = \frac{\pi}{4} \rho C_m D_i^2 L_i \int_0^l \frac{dv^n(x)}{dt} dx \quad (3)$$

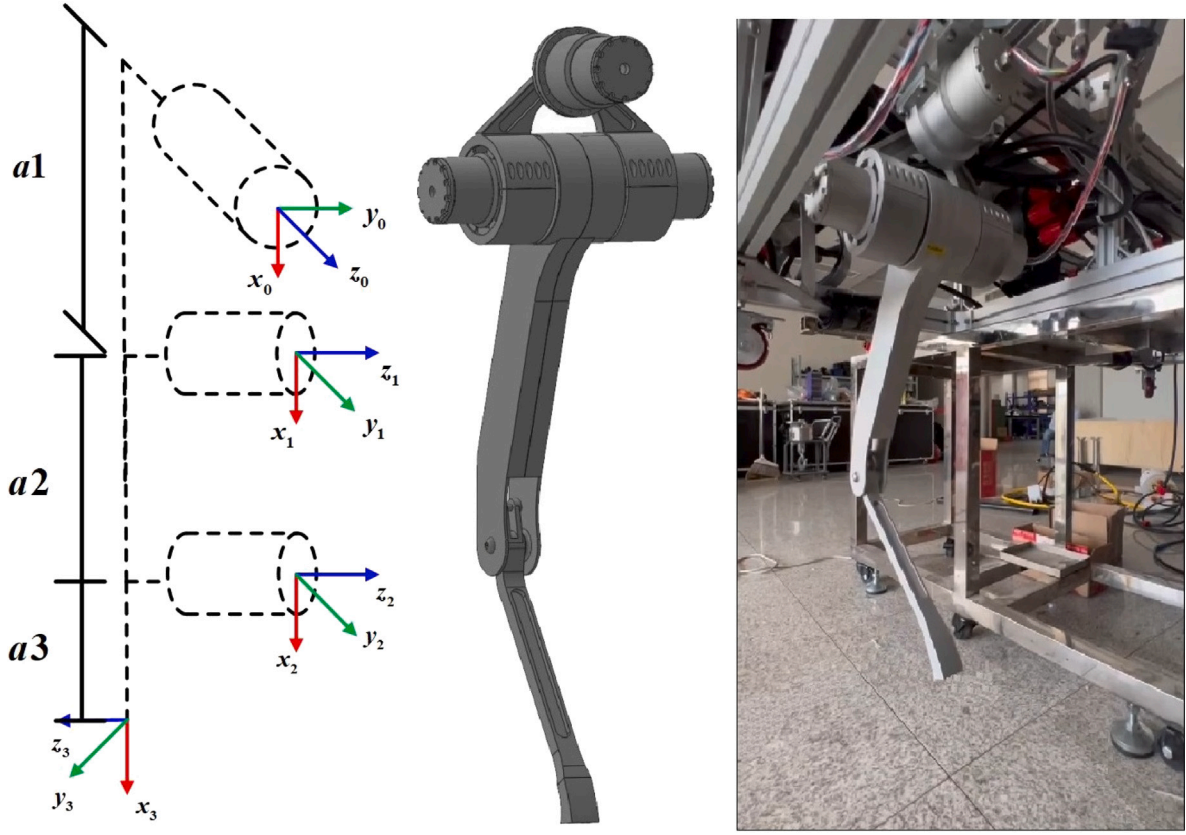


Fig. 1. The underwater manipulator.

where ρ represents the density of water, D_i represents the diameter of the i th cylindrical rod, L_i represents the moment arm of the i th joint, $v^n(x)$ is the velocity vector in the normal direction of the cylindrical surface, C_r represents the water resistance coefficient, and C_m represents the additional mass force coefficient.

Let $x = [q^T, \dot{q}^T]^T$ be the state vector, $u = \tau$ be the control input, so that the dynamics model of the underwater manipulator can be rewritten as,

$$\dot{x} = f(x, u) + w \quad (4)$$

where

$$f(x, u) = \begin{bmatrix} \dot{q} \\ M^{-1}(q)[\tau - C(q, \dot{q}) - G(q) - F(q, \dot{q})] \end{bmatrix} \quad (5)$$

$$w = \begin{bmatrix} 0 \\ I \end{bmatrix} \tau_d \quad (6)$$

To linearize the underwater manipulator's nonlinear dynamics in discrete time, we can use a Taylor series expansion around an operating point. The model can be approximated as,

$$x_{k+1} = Ax_k + Bu_k + w_k \quad (7)$$

where $x_k = x - x_r \in \mathbb{R}^{2n}$, $u_k = u - u_r \in \mathbb{R}^n$, the linearization matrices A and B are obtained from the Jacobian of the system equations: $A = I + \delta_t \frac{\partial f}{\partial x} \Big|_{(x_r, u_r)}$, $B = \delta_t \frac{\partial f}{\partial u} \Big|_{(x_r, u_r)}$. Define q_d and $\dot{q}_d$ as the tracking reference for the joint position and joint velocity, then let $x_r = [q_d^T, \dot{q}_d^T]^T$ be the reference state vector, and u_r is the reference control input, δ_t is the sampling time.

In this paper, the reference control input $u_r = 0$ is set. In analyzing the underwater manipulator system, the following two assumptions are proposed.

Assumption 1. The system state variables x_k and control variables u_k are subject to general polyhedral constraints at any time step k , as specified by

$$(x_k, u_k) \in \gamma := \{(x_k, u_k) : \forall i \in \mathbb{N}_{[1:p]}, c_i^T x_k + d_i^T u_k \leq 1\} \quad (8)$$

This means that the state variables x_k and control variables u_k are confined within a polyhedral region, which can be described by a set of linear inequalities (8). Each pair (c_i, d_i) is known exactly, and the number of constraints p is finite.

Assumption 2. The external disturbance variables w_k are bounded and subject to general polyhedral constraints at any time step k , as specified by

$$w_k \in W := \{w \in \mathbb{R}^n : \forall i \in \mathbb{N}_{[1:l]}, \omega_i^T w \leq 1\} \quad (9)$$

This means that the disturbance variables w_k are confined within a polyhedral region, which can be described by a set of linear inequalities (9), where ω_i is known exactly, and the number of constraints l is finite.

In this paper, a robust MPC scheme is designed for the motion control of a newly designed underwater manipulator with three degrees of freedom. The proposed IRTMPC-ASMC scheme aims to overcome bounded matched lumped disturbances, which include unknown unmodeled uncertainties and time-varying external disturbances.

4. Inertial parameter identification of the underwater manipulator

Before controlling an actual manipulator, it is essential to consider that various practical factors, such as machining processes and the assembly of mechanical parts, can cause the dynamic parameters of the manipulator to differ from those measured in simulations. These

variations can directly impact the accuracy of the control system. Therefore, it is necessary to perform a preliminary estimation of the inertial parameters of the underwater manipulator. The inertial parameters of the underwater manipulator studied in this section include mass, moments of inertia, and the center of mass. The least squares method will be used to estimate these unknown inertial parameters.

To estimate the inertial parameters of the underwater manipulator, it is necessary to rewrite the dynamic equation (1) of the manipulator into a linear form of the inertial parameters, as shown in equation (10).

$$\tau = \varphi(q, \dot{q}, \ddot{q}, S_0)X(\Theta_I) \quad (10)$$

where $\varphi(q, \dot{q}, \ddot{q})$ represents a matrix that includes joint positions, velocities, accelerations, and the state of the root joint $S_0 = [v_b, \omega_b, \dot{v}_b, \dot{\omega}_b]$, $v_b, \omega_b, \dot{v}_b, \dot{\omega}_b$ denote the angular velocity, angular acceleration, linear velocity, and linear acceleration of the root joint, respectively. It should be noted that the root joint of the robotic arm remains stationary during the identification process $S_0 = 0$. τ is a matrix representing the output torques at the joints, and $X(\Theta_I)$ contains the manipulator's inertia parameters, referred to as the minimum inertia parameter set.

The inertia parameters for each link of the manipulator, including moments of inertia, center of mass, mass, sliding friction, and viscous friction.

$$\Theta_I = [I_{xx}^i, I_{xy}^i, I_{xz}^i, I_{yz}^i, I_{zz}^i, I_{x'}^i, I_{y'}^i, I_{z'}^i, m^i, f_r^i, f_v^i] \quad (11)$$

$\varphi(q, \dot{q}, \ddot{q}, 0)$ and τ can be obtained through experimental measurements. Using these matrices, the minimal inertia parameter set $X(\Theta_I)$ can be estimated using the least squares method. The specific form of the least squares method is shown in Table 2.

4.1. Excitation functions optimization

In the process of parameter identification for a robotic manipulator, the first step is to determine the excitation function. This function is used to control the manipulator to follow a desired position tracking curve. This motion allows us to obtain the time-varying joint positions, velocities, accelerations, and the output torques required for identification. However, the choice of the excitation function directly affects the accuracy of the identification. Therefore, it is necessary to optimize the excitation function. The excitation function can be expressed as follows,

$$q_i(t) = \sum_{l=1}^n (a_l \sin(w_l t) - b_l \cos(w_l t)) + q_i(0) \quad (12)$$

The parameter selection for the excitation functions is performed using an optimization algorithm based on an improved genetic algorithm. The workflow is described in Algorithm 3. Specific calculation details can be found in Xu et al. (2022): here, it is necessary to explain the following:

- Initialize parameters: Maximum number of iterations N_{max} , maximum fitness F_{max} , annealing coefficient K , initial degradation temperature t_0 , and mutation rate λ .
- The population $IP = [a_i, b_i, w_i, q_i(0)]$.
- The fitness function $F = \frac{\max_{SV}(\varphi(q, \dot{q}, \ddot{q}))}{\min_{SV}(\varphi(q, \dot{q}, \ddot{q}))}$.
- Selection algorithm: The monarch selection method is adopted, that is, after the population fitness is sorted, the individual with the optimal fitness and the even-numbered individuals are crossed.
- Crossover algorithm: Two-point crossover is used, where two elements are randomly exchanged in IP .
- Mutation algorithm: Single-point mutation is used, where one element is randomly mutated in IP .

Algorithm 1 Parameter Optimization of Excitation Functions for Parameter Identification Using an Improved Genetic Algorithm (N_{max} , F_{max} , K , t_0 , λ)

```

1: Initialize the population  $IP$ 
2: while  $N < N_{max}$  and  $F < F_{max}$  do
3:   Compute the fitness function  $F(IP)$ 
4:   Preserve the best fit individuals  $IP_{prof}(IP, F)$ 
5:   Perform crossover and mutation operations  $IP'(IP, K, t_0, \lambda)$ 
6:   Eliminate similar individuals based on niche exclusion
        $IP_{new}(IP_{prof}, IP')$ 
7:   if the variance of population genes  $VF < 1$  then
8:     Exclude individuals with the lowest fitness  $IP(IP_{new})$ 
9:   else
10:     $IP = IP_{new}$ 
11:   end if
12:   Update parameters based on simulated annealing algorithm
13: end while
14: return  $IP$ 

```

4.2. Least squares identification based on integral operators

Since joint accelerations $\ddot{q}$ cannot be directly measured by sensors, simply performing numerical differentiation on joint velocities would result in significant errors. Therefore, this paper adopts the least squares identification method based on integral operators. This approach avoids the appearance of acceleration terms in the regression model and thus improves the identification accuracy.

First, rewrite Eq. (10) as:

$$\tau(t) = \left[\frac{d}{dt} \varphi_1(t) + \varphi_0(t) \right] X \quad (13)$$

and then perform the Laplace transform on both sides of Eq. (13),

$$\tau(s) = [s\varphi_1(s) - \varphi_1(0) + \varphi_0(s)] X \quad (14)$$

Then, multiply both sides of Eq. (14) by a first-order low-pass filter $\frac{\lambda}{s+\lambda}$.

$$\begin{aligned} & \frac{\lambda}{s+\lambda} \tau(s) \\ &= \frac{\lambda}{s+\lambda} [s\varphi_1(s) - \varphi_1(0) + \varphi_0(s)] X \\ &= \left[\lambda\varphi_1(s) - \frac{\lambda^2}{s+\lambda} \varphi_1(s) - \frac{\lambda}{s+\lambda} \varphi_1(0) + \frac{\lambda}{s+\lambda} \varphi_0(s) \right] X \\ &= \left[\lambda\varphi_1(s) + y_1(s) + \frac{\lambda}{s+\lambda} \varphi_0(s) \right] X \end{aligned} \quad (15)$$

Then, perform the inverse Laplace transform on the resulting equation to obtain

$$\tau^{1f}(t) = \varphi^{1f}(t)X \quad (16)$$

where

$$\tau^{1f}(t) = L^{-1} \left\{ \frac{\lambda}{s+\lambda} \tau(s) \right\} \quad (17)$$

$$\begin{aligned} \varphi^{1f}(t) &= L^{-1} \left\{ \lambda\varphi_1(s) + y_1(s) + \frac{\lambda}{s+\lambda} \varphi_0(s) \right\} \\ &= \lambda\varphi_1(t) + y_1(t) + \varphi_0^{1f}(t) \end{aligned} \quad (18)$$

The components containing integral operators, $\tau^{1f}(t)$, $y_1(t)$ and $\varphi_0^{1f}(t)$, can be computed through numerical integration using the differential equations in Eq. (19).

$$\begin{cases} \dot{\tau}^{1f}(t) = -\lambda\tau^{1f}(t) + \lambda\tau(t), \tau^{1f}(0) = 0 \\ \dot{y}_1(t) = -\lambda y_1(t) - \lambda^2 \varphi_1(t), y_1(0) = -\lambda\varphi_1(0) \\ \dot{\varphi}_0^{1f}(t) = -\lambda\varphi_0^{1f}(t) + \lambda\varphi_0(t), \varphi_0^{1f}(0) = 0 \end{cases} \quad (19)$$

Finally, the minimal inertia parameter set can be estimated using Eq. (20). The complete algorithmic process is presented in Algorithm 2.

$$\hat{X} = \left[(\varphi^{1f})^T \varphi^{1f} \right]^{-1} (\varphi^{1f})^T \tau^{1f}(t) \quad (20)$$

Algorithm 2 Least Squares Identification Based on Integral Operators

- 1: Perform QR decomposition on the dynamics model of the manipulator to obtain the regression equation.
 - 2: Design the excitation function $q_i(t)$.
 - 3: Optimize the excitation function using a genetic algorithm to obtain the optimal excitation function $q_i^*(t)$.
 - 4: Rewrite the regression equation with acceleration term and perform Laplace transform $\tau(s)$.
 - 5: Introduce a first-order low-pass filter $\frac{\lambda}{s+\lambda}$ and perform the inverse Laplace transform to obtain $\tau^{1f}(t) = \varphi^{1f}(t)X$.
 - 6: Calculating Components with Integral Operators $\tau^{1f}(t)$, $y_1(t)$, and $\varphi_0^{1f}(t)$.
 - 7: Computing the minimal inertial parameter set $\hat{X}$.
 - 8: **return** $\hat{X}$.
-

5. Implicit rigid tube robust model predictive control scheme with adaptive sliding mode strategy

Due to inevitable errors in the parameter identification process of underwater manipulators and the influence of external factors such as underwater currents during actual operation, these factors jointly affect the control accuracy and stability of underwater manipulators. To enhance the robustness of underwater manipulator control and better address these challenges, this paper adopts a tube-based robust model predictive control scheme. By constructing a predictive model and considering uncertainty boundaries, this scheme achieves more stable and precise control even in the presence of parameter identification errors and external disturbances.

Especially for high-dimensional systems, traditional methods require extensive set operations when calculating the explicit minimal robust set, which not only increases computational complexity but may also affect the real-time performance of control. To overcome this difficulty, this paper innovatively employs a combination of implicit rigid tube MPC and adaptive sliding mode methods. This approach effectively reduces the computational load while maintaining control robustness, enabling rapid and stable motion control of underwater manipulators. To more intuitively demonstrate the implementation process of this control strategy, this paper provides a control block diagram, as shown in Fig. 2.

The IRTMPC-ASMC decomposes into a coordinated strategy combining traditional MPC and an adaptive sliding mode auxiliary controller. The proposed controller's form can be expressed as,

$$u_k = v_k + \Phi e_k \quad (21)$$

where v_k represents the nominal MPC control law, and Φe_k represents the adaptive sliding mode control law, $e_k = x_k - z_k$ represents the difference between the actual state and the nominal state of the system. The detailed calculation process will be introduced in subsequent sections.

This approach ensures both state variable and control input constraints while effectively enhancing the robustness of the algorithm. IRTMPC-ASMC uses the auxiliary controller to keep the system state within a robust invariant set, maintaining bounded motion around the nominal control state of the traditional MPC.

5.1. Implicit rigid tube model predictive control

IRTMPC is an iterative decision-making process where the primary task is to determine the optimal control of the implicit rigid tube. In this approach, the nominal model of the system (22) is first used to forecast behavior under uncertainty. This prediction is made using implicit rigid state tubes and corresponding affine control strategies.

$$z_{k+1} = Az_k + Bv_k \quad (22)$$

In this context, z_k represents the nominal state variable, and the optimal control for the nominal system is formulated as the following optimization problem,

$$J = \min_{u_k} \sum_{i=1}^{N-1} \left(\|z_{k+i}\|_Q^2 + \|v_{k+i}\|_R^2 \right) + \|z_{k+N}\|_P^2 \quad (23)$$

With respect to $\{z_k\}_{k=0}^N$ and $\{v_k\}_{k=0}^N$

Subject to

$$\forall k \in \mathbb{N}_{[1:N-1]}, z_{k+1} = Az_k + Bv_k$$

$$\forall k \in \mathbb{N}_{[1:N-1]}, (z_k, v_k) \in y_s$$

$$z_{N+1} = (A + BK)z_N$$

$$z_N \in z_f$$

where y_s , z_f represent the stage constraint set and the terminal constraint set, respectively. K is a positive definite matrix that ensures that the closed-loop system $A + BK$ is strictly stable, and K is calculated offline by the infinite horizon linear quadratic regulator (LQR), as specified by

$$K = [I, 0, \dots, 0] (\phi^T Q_c \phi + R_c)^{-1} \phi^T Q_c F \quad (24)$$

Where the terminal gain matrix P is solved by the Riccati equation,

$$P = (A + BK)^T P (A + BK) + (Q + K^T R K) \quad (25)$$

Theorem 1. *If the constrained optimization problem (23) has a solution at the initial time, and the selections of K and P are given by Equations (24) and (25) respectively, then the constrained optimization problem (23) has a solution at any time larger than 0, and the closed-loop system is asymptotically stable.*

Proof. The detailed proof is the same as the paper in Liu et al. (2023).

5.2. The constraints of implicit rigid tube model predictive control

Due to the introduction of the auxiliary control rate Φ , to ensure the constraints of the closed-loop system (Assumption 1), y_s will be a more compact constraint compared to y .

Substituting the control rate (21) into the system state equation (7) yields,

$$x_{k+1} = Ax_k + Bv_k + B\Phi(x_k - z_k) + w_k \quad (26)$$

Subtracting the nominal model in equation (22), from equation (26), yields the error model between the actual system and the nominal system,

$$\begin{aligned} e_{k+1} &= (A + B\Phi)e_k + w_k \\ e_{k+2} &= (A + B\Phi)^2 e_k + (A + B\Phi)w_k + w_{k+1} \\ &\vdots \\ e_{k+m} &= (A + B\Phi)^m e_k + \sum_{i=0}^{m-1} [(A + B\Phi)^{m-1-i} w_{k+i}] \end{aligned} \quad (27)$$

From the above recurrence relation, e_k can be expressed as,

$$e_k = (A + B\Phi)^k e_0 + \sum_{i=0}^{k-1} [(A + B\Phi)^{k-1-i} w_{0+i}] \quad (28)$$

Assumption 3. At the initial time $k = 0$, the actual state of the system is equal to the nominal state, as specified by,

$$e_0 = x_0 - z_0 = \vec{0} \quad (29)$$

Eq. (28) can be simplified to,

$$e_k \in \bigoplus_{i=0}^{k-1} (A + B\Phi)^i W \quad (30)$$

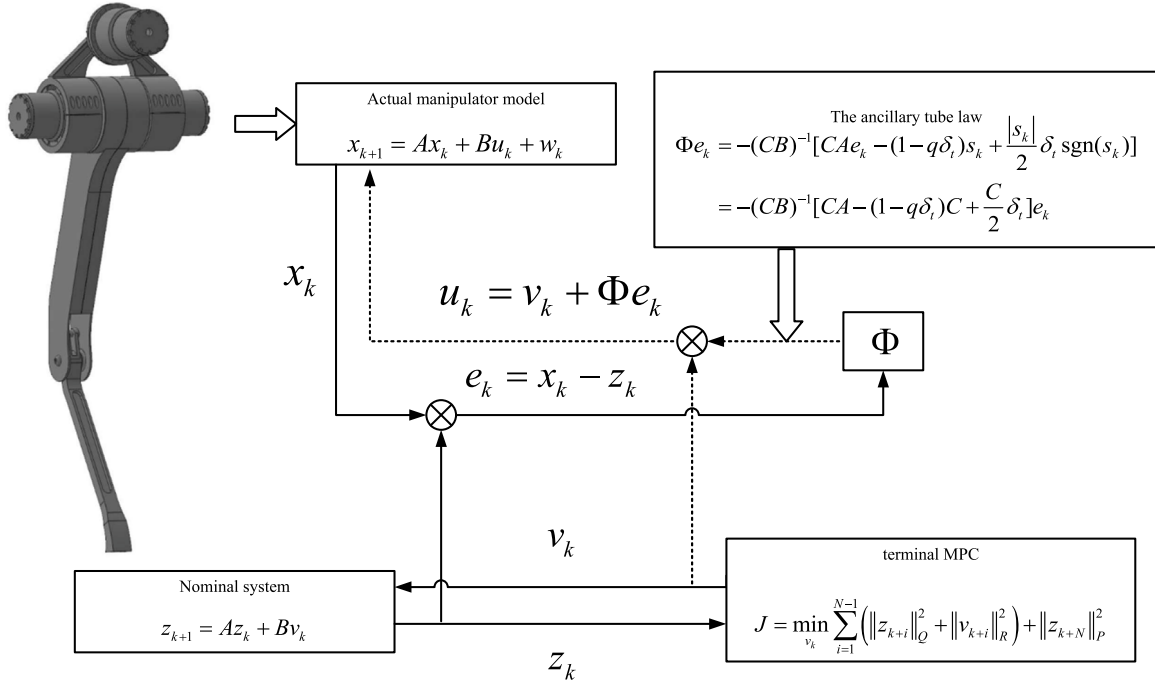


Fig. 2. The workflow of the proposed IRTMPC-ASMC scheme.

Theorem 2 (Mayne et al., 2005). When the matrix Φ is known exactly and such that the matrix $A + B\Phi$ is strictly stable (all the eigenvalues of $A + B\Phi$ are strictly inside the unit disk), we can obtain the following results,

1. There exists a scalar $\alpha \in [0, 1)$ and a finite integer $N_S \in \mathbb{N}$ satisfy $(A + B\Phi)^{N_S} W \subseteq \alpha W$.
2. The minimal positive robust set of the system is $S = (1 - \alpha)^{-1} \bigoplus_{i=0}^{N_S-1} (A + B\Phi)^i W$. That is, for all $e_k \in S$ and all $w \in W$, $e_{k+1} \in S$.

Here is the computation process for determining N_S and its corresponding α .

Algorithm 3 Offline computation process of parameters for the minimal positive robust set $(A + B\Phi, W, \epsilon)$

- 1: Initialize N_S, α
- 2: **while** $\alpha > \frac{\epsilon}{\epsilon + \max \left\{ \sum_{k=1}^{N_S} h(W, (A + B\Phi)^k) \right\}}$ **do**
- 3: $\alpha = \max \left\{ h(W, (A + B\Phi)^k \omega_i) \right\}$
- 4: $N_S = N_S + 1$
- 5: **end while**
- 6: **return** N_S, α

From the above theorem, the following two corollaries can be derived,

1. For $\forall s \in S$, $(s, \Phi s) \in y$. Substituting this corollary into Eq. (8) yields

$$\forall s \in S, c_i^T s + d_i^T \Phi s < 1 \quad (31)$$

which can be expressed as,

$$h(S, c_i + \Phi^T d_i) < 1 \quad (32)$$

2. The constraint Eq. (8) of the system can be rewritten as,

$$y := \{(z_k, v_k) : \forall i \in \mathbb{N}_{[1:p]}, \forall s \in S, c_i^T(z_k + s) + d_i^T(v_k + \Phi s) \leq 1\} \quad (33)$$

According to Eq. (32), The constraint relationship of the auxiliary control rate can be obtained as,

$$(c_i^T + d_i^T \Phi) s \leq h(S, c_i + \Phi^T d_i) < 1, \forall s \in S \quad (34)$$

Furthermore, the tighter stage constraints, also known as the nominal system constraints, can be represented as follows:

$$y_s := \{(z_k, v_k) : \forall i \in \mathbb{N}_{[1:p]}, c_i^T z_k + d_i^T v_k \leq 1 - h(S, c_i + \Phi^T d_i)\} \quad (35)$$

So far, the stage constraint y_s for the nominal state and output torque in optimal control has been determined. However, when calculating the minimal positive robust set S , a large number of high-dimensional Minkowski sums are required, which will consume a significant amount of system resources and seriously affect the real-time performance of the system.

Theorem 3 (Raković, 2023). Let Ω be a closed convex subset containing the origin in $\mathbb{R}^n$, $M \in \mathbb{R}^{n \times n}$, $k \in \mathbb{N}$. For all $y \in \mathbb{R}^n$, there exists

$$h(\bigoplus_{i=0}^k M^i \Omega, y) = \sum_{i=0}^k h(\Omega, (M^i)^T y) \quad (36)$$

Theorem 3 transforms a series of complex set operations into simple summation operations, avoiding the explicit solution of the minimal positive robust set. The stage constraint (35) can be further represented as,

$$y_s = \{(z, v) : \forall i \in \mathbb{N}_{[1:p]}, c_i^T z + d_i^T v \leq 1 - f_i\} \quad (37)$$

where

$$\begin{aligned} f_i &= h(S, c_i + \Phi^T d_i) \\ &= h((1 - \alpha)^{-1} \bigoplus_{i=0}^{N_S-1} (A + B\Phi)^i W, c_i + \Phi^T d_i) \\ &= (1 - \alpha)^{-1} \sum_{j=0}^{N_S-1} h(W, ((A + B\Phi)^j)^T (c_i + \Phi^T d_i)) \end{aligned} \quad (38)$$

By subtracting the nominal MPC constraint (8) from the total system constraint (35), the constraint imposed by the auxiliary ASMC law can be obtained,

$$\begin{aligned} E_s &= \{e_k\} : \forall i \in \mathbb{N}_{[1:p]}, \\ (c_i^T + d_i^T \Phi) e_k &\leq h(S, c_i + \Phi^T d_i) \end{aligned} \quad (39)$$

Then, the terminal constraint set can be represented as:

$$z_f = \{(z_N, K_z z_N) : \forall i \in \mathbb{N}_{[1:p]}, (c_i^T + d_i^T K_z) z_N \leq 1 - f_i\} \quad (40)$$

5.3. Auxiliary adaptive sliding mode control law

The adaptive sliding mode control law designed in this paper will make the trajectory of the error system centered along the nominal trajectory z_k within the invariant set S . The design of the auxiliary adaptive sliding mode control law is as follows:

$$\begin{aligned} \Phi e_k &= -(CB)^{-1} [CAe_k - (1 - q\delta_t)s_k + \frac{|s_k|}{2}\delta_t \operatorname{sgn}(s_k)] \\ &= -(CB)^{-1} [CA - (1 - q\delta_t)C + \frac{C}{2}\delta_t]e_k \end{aligned} \quad (41)$$

The design of the sliding mode surface is as follows:

$$s_k = Ce_k \quad (42)$$

where $C \in \mathbb{R}^{n \times 2n}$ is a constant matrix, and the sliding mode reaching law can be expressed as:

$$s_{k+1} = s_k - q\delta_t s_k - \frac{|s_k|}{2}\delta_t \operatorname{sgn}(s_k) \quad (43)$$

Theorem 4. For the discrete-time error system (27), applying the auxiliary adaptive sliding mode control law in Eq. (41) can ensure that the system matrix $A + BK$ is strictly stable.

Proof. Choose the Lyapunov function as:

$$V = s_k^2 \quad (44)$$

Substituting the sliding mode reaching law (43) into the function yields:

$$\begin{aligned} \Delta V &= s_{k+1}^2 - s_k^2 \\ &= (s_{k+1} + s_k) \operatorname{sgn}(s_k)(s_{k+1} - s_k) \operatorname{sgn}(s_k) \\ &= [(2 - q\delta_t)s_k - \frac{|s_k|}{2}\delta_t \operatorname{sgn}(s_k)] \operatorname{sgn}(s_k) \\ &\quad [-q\delta_t s_k - \frac{|s_k|}{2}\delta_t \operatorname{sgn}(s_k)] \operatorname{sgn}(s_k) \\ &= (2 - q\delta_t - 0.5\delta_t) |s_k| [- (q + 0.5)\delta_t] |s_k| \end{aligned}$$

Therefore, when the sampling time satisfies $\delta_t < \frac{4}{1+2q}$ such that $\Delta V < 0$, the sliding mode reaching law meets the existence and reaching conditions of discrete sliding mode control. The designed control system is stable, and the matrix $A + BK$ is strictly stable. Then, according to Theorem 2, the control output will satisfy the stage state constraint (33).

6. Experiment and analysis

This section considers the three-degree-of-freedom underwater manipulator shown in Fig. 1 and the D-H parameter of the manipulator are given in Table 1. The distinctive feature of this underwater manipulator lies in its structural and functional optimization. It is equipped with three identical drive motors, all of which are servo motors with harmonic reducers, ensuring efficient and stable power output. Each motor's output shaft is specially fitted with dual layers of watertight O-rings, enabling the underwater manipulator to operate normally at a depth of 300 meters underwater, demonstrating exceptional underwater operational capability.

Meanwhile, the motors receive control signals in the form of output torques, which are transmitted from the control computer via a CAN bus at a frequency of up to 1000 Hz. Simultaneously, the motors return their operational data to the control computer at the same frequency, including current, velocity, and position information. This bidirectional communication ensures precise control and real-time monitoring of the underwater manipulator.

Table 1

The D-H parameter of the manipulator.

Link	a_i (m)	α_i (°)	d_i (m)	θ (°)	D_i (m)
l_1	0.15	-90	0	q_1	0.145
l_2	0.52	0	0	q_2	0.051
l_3	0.42	180	0	q_3	0.03

Table 2

The minimal inertia parameter set $X(\Theta_i)$.

$X(\Theta_i)$	Identification results
$I_{1yy} + I_{2yy} + I_{3yy} - 0.25L_3m_1 - 2L_1m_2 - L_2m_3$	4.31195
$l_{1x} + L_1m_1 + L_1m_2 + L_1m_3$	0.09433
$l_{1z} + l_{2z} - l_{3z}$	0.01913
f_{1v}	5.85504
f_{1r}	0.61678
$I_{2xx} - I_{2yy} + 2L_2m_2 + 2L_2m_3$	-6.55937
I_{2xy}	0.17692
$I_{2xz} - L_2l_{2z} + L_2l_{3z}$	-0.19130
I_{2yz}	-1.77409
$I_{2zz} - 2L_2m_2 - 2L_2m_3$	0.38672
$l_{2x} - L_2m_2 + L_2m_3$	0.09322
l_{2y}	-0.01177
f_{2v}	1.11485
f_{2r}	0.55473
$I_{3xx} - I_{3yy} + 0.3L_2m_3$	-0.67309
I_{3xy}	-0.05098
$I_{3xz} - L_3l_{3z}$	-0.06071
I_{3yz}	0.01565
$I_{3zz} - 0.3L_2m_3$	0.07919
$l_{3x} + L_3m_3$	0.01674
l_{3y}	0.01239
f_{3v}	1.10677
f_{3r}	0.50489

The rest of this section will conduct simulation and field experiments to testify and analyze the performance of the presented parameter identification method and the proposed IRTMP-ASMC scheme in the trajectory control of the underwater manipulator.

6.1. Inertia parameter identification

First, this section conducts an experiment to estimate the inertia parameters of the underwater manipulator. By using the least squares identification method based on integral operators proposed in Section 4, the inertia parameters in the dynamic model of the underwater manipulator are identified for subsequent simulation and field experiments. The specific process is as follows:

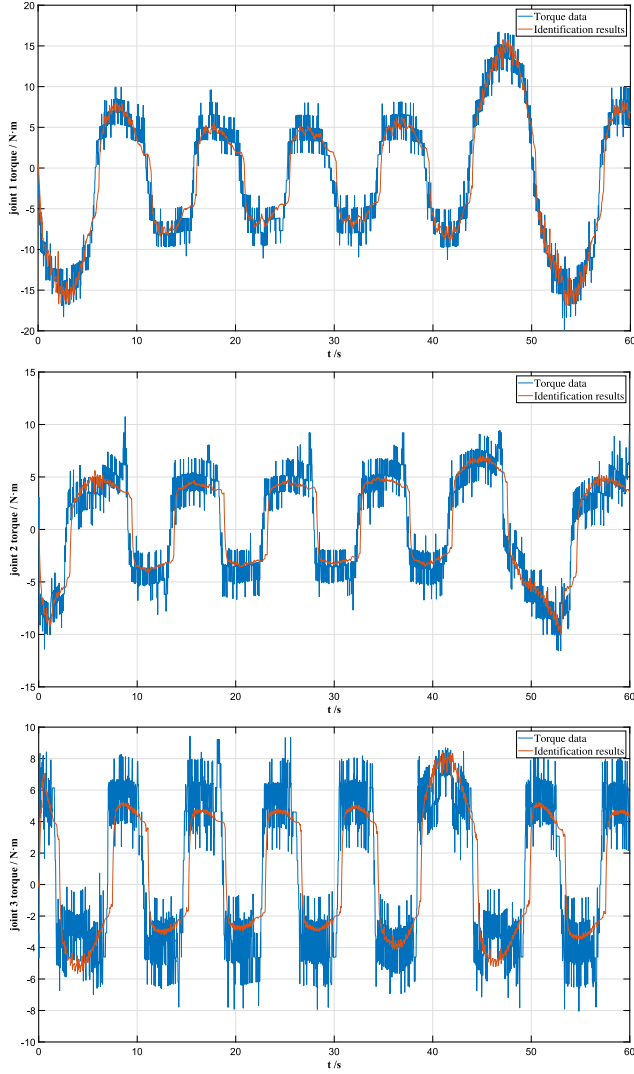
- Step 1. Optimize the parameters in the excitation function (12). Using Algorithm 3 introduced in Section 4.1, the trajectory parameters are obtained as shown in Table 3.
- Step 2. Conduct trajectory tracking control of the underwater manipulator on the surface using the PID, and record the positions, speeds, and torques of the joints during the motion.
- Step 3. Identify the inertia parameters of the underwater manipulator. By processing the motion data through Algorithm 2 introduced in Section 4.2, the inertia parameters are obtained as shown in Table 2.
- Step 4. Substitute the obtained inertia parameters into Eq. (10). The identified output torque is compared with the actual torque to verify the feasibility of the algorithm.

The estimated output torque and the actually measured torque are shown in Fig. 3. It can be seen through comparison that the estimated torque matched with the actual measured torque, which strongly proves the effectiveness and accuracy of the identified parameters and the identification method.

Table 3

The parameter of the excitation functions.

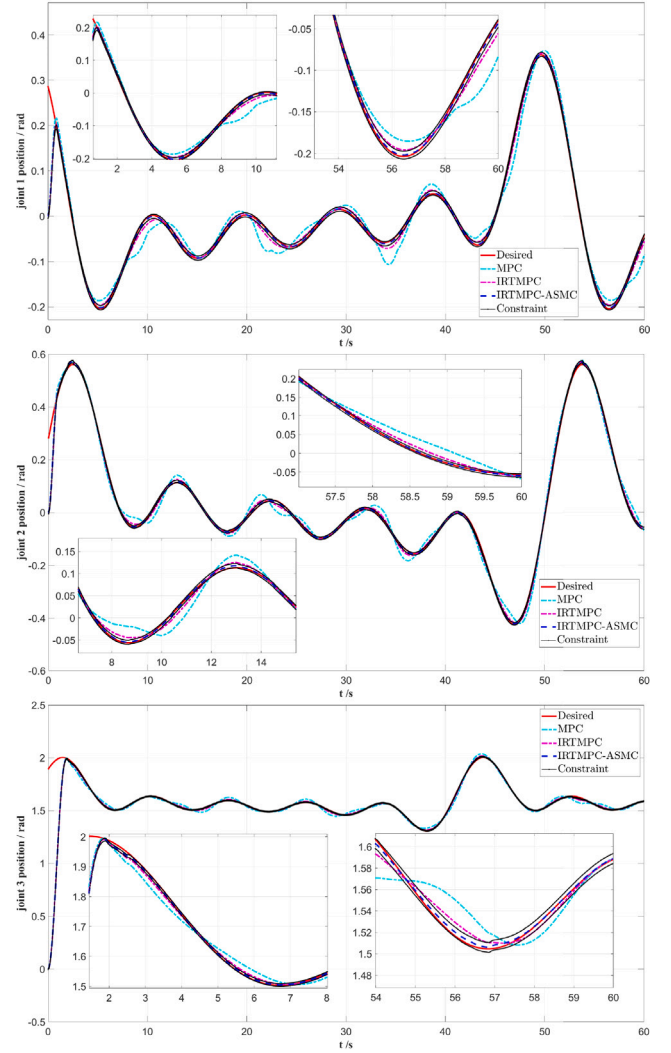
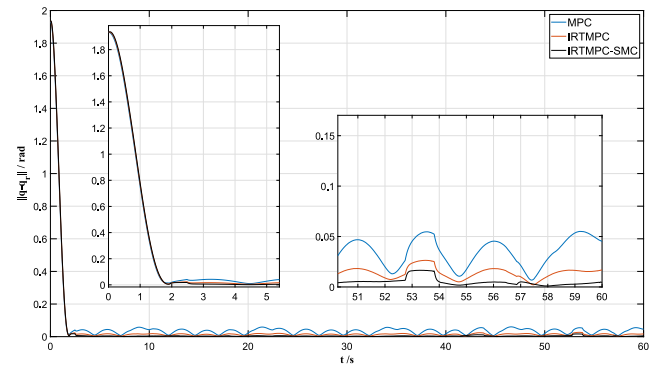
Joint	a_i	b_i	w_i	$q_i(0)$
J_1	-0.04725	0.05740	0.1229	0
J_2	-0.1127	-0.05588	0.1224	0
J_3	0.06539	0.06391	0.14836	1.571

**Fig. 3.** The estimated torque and the actual torque.

6.2. Simulations

This section verified the feasibility of the proposed IRTMPC-ASMC scheme in the trajectory tracking control of the manipulator through simulation experiments. The desired trajectory is used the excitation function (12), with its parameters detailed in Table 3. The performance of the proposed IRTMPC-ASMC scheme has also been compared with that of the traditional MPC and IRTMPC in the simulation experiments. It is particularly noteworthy that the feedback gain matrix Φ in the IRTMPC method used for comparison is obtained by calculating the infinite horizon LQR of J (24).

To comprehensively demonstrate the control effectiveness of the proposed IRTMPC-ASMC scheme under various types of internal and external disturbances, we designed the disturbance $w = \delta + w_u + w_d$ to include three main components. First, δ represents system measurement noise, specifically modeled as Gaussian white noise with a mean of

**Fig. 4.** The comparison of tracking by manipulator joints.**Fig. 5.** The comparison of tracking error by manipulator joints.

0 and a variance of 0.001. Second, $w_u = A(1 + 10\%)x_k$ represents model uncertainty, assuming that unmodeled uncertainty affects 10%. Finally, w_d represents external disturbances during operation, with its specific form expressed as Eq. (45). Through this design, we can

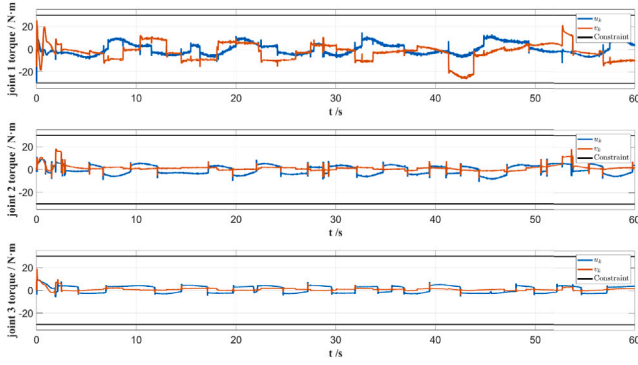


Fig. 6. The output torque by manipulator joints.

more thoroughly evaluate the control performance of the proposed IRTMPC-ASMC scheme in various complex environments.

$$w_d = \delta_t \begin{bmatrix} 0.08 \sin(0.05t) + 0.1 \cos(0.01t) \\ 0.15 \sin(0.01t) + 0.09 \cos(0.015t) \\ 0.1 \sin(0.015t) + 0.07 \cos(0.01t) \\ 0.8 \sin(0.03t) + 0.5 \cos(0.04t) \\ 0.6 \sin(0.04t) + 0.6 \cos(0.03t) \\ 0.3 \sin(0.05t) + 0.4 \cos(0.03t) \end{bmatrix} \quad (45)$$

To simulate the underwater environment, the fluid resistance torque τ_r and the added mass torque τ_m are introduced into the output torque u_k , treating these two torques as part of the unknown disturbances. Their specific expressions are given as follows:

The water resistance torque on the joints,

$$\begin{aligned} T_{d1} &= \frac{1}{2} \rho C_d D_1 \int_0^{a_1} \dot{\theta}_1 x^2 \|\dot{\theta}_1 x\| dx \\ &\quad + \frac{1}{2} \rho C_d D_2 \int_0^{a_2} \dot{\theta}_1 \gamma_{12}^2 \|\dot{\theta}_1 \gamma_{12}\| dx \\ &\quad + \frac{1}{2} \rho C_d D_3 \int_0^{a_3} \dot{\theta}_1 \gamma_{13}^2 \|\dot{\theta}_1 \gamma_{13}\| dx \\ T_{d2} &= \frac{1}{2} \rho C_d D_2 \int_0^{a_2} \dot{\theta}_2 x^2 \|\dot{\theta}_2 x\| dx \\ &\quad + \frac{1}{2} \rho C_d D_3 \int_0^{a_3} \cos \beta \dot{\theta}_2 \gamma_{23}^2 \|\dot{\theta}_2 \gamma_{23}\| dx \\ &\quad + \cos \beta \dot{\theta}_2 \gamma_{23} \dot{\theta}_3 x \|\dot{\theta}_3 x\| dx \\ T_{d3} &= \frac{1}{2} \rho C_d D_3 \int_0^{a_3} x \dot{\theta}_3 x \|\dot{\theta}_3 x\| + x \cos \beta \dot{\theta}_2 \gamma_{23} \|\dot{\theta}_2 \gamma_{23}\| dx \end{aligned}$$

The additional mass torque on the joints,

$$\begin{aligned} T_{m1} &= \frac{\pi}{4} \rho C_m D_1^2 \int_0^{a_1} \ddot{\theta}_1 x^2 dx \\ &\quad + \frac{\pi}{4} \rho C_m D_2^2 \int_0^{a_2} \ddot{\theta}_1 \gamma_{12}^2 + \dot{\theta}_1 \gamma_{12} \gamma_{13} dx \\ &\quad + \frac{\pi}{4} \rho C_m D_3^2 \int_0^{a_3} \ddot{\theta}_1 \gamma_{13}^2 + \dot{\theta}_1 \gamma_{13} \gamma_{13} dx \\ T_{m2} &= \frac{\pi}{4} \rho C_m D_2^2 \int_0^{a_2} \ddot{\theta}_2 x^2 dx \\ &\quad + \frac{\pi}{4} \rho C_m D_3^2 \int_0^{a_3} \cos \beta r_{23} (\ddot{\theta}_2 \gamma_{23} + \dot{\theta}_2 \dot{\gamma}_{23} + \ddot{\theta}_3 x) dx \\ T_{m3} &= \frac{\pi}{4} \rho C_m D_3^2 \int_0^{a_3} \ddot{\theta}_3 x^2 + x \cos \beta (\ddot{\theta}_2 \gamma_{23} + \dot{\theta}_2 \dot{\gamma}_{23}) dx \end{aligned}$$

where,

$$\gamma_{12} = (x + a_1) \cos \theta_2$$

$$\gamma_{13} = (a_1 + a_2 \cos \theta_2 + x \cos(\theta_2 + \theta_3))$$

$$\gamma_{23} = \sqrt{a_2^2 + x^2 + 2a_2 x \cos \theta_3}$$

$$\cos \beta = \frac{a_2^2 + \gamma_{23}^2 - x^2}{2a_2 \gamma_{23}}$$

The values of C_r and C_m , which are used in this simulation are obtained from empirical values $C_r = 1.0$ and $C_m = 2.0$ (Yuguang and Fan, 2019).

To simulate the impact of the velocity and acceleration of the underwater vehicle on the control performance of the manipulator, the torques τ_{fd} is introduced into the output torque u_k , treating τ_{fd} as part of the unknown disturbances.

$$\begin{aligned} \tau_{fd} &= \varphi(q, \dot{q}, \ddot{q}, S_0) X(\Theta_i) - \varphi(q, \dot{q}, \ddot{q}, 0) X(\Theta_i) \\ &= f_d(q, \dot{q}, \ddot{q}, v_b, w_b, \dot{v}_b, \dot{w}_b) \end{aligned} \quad (46)$$

where the following trigonometric functions are used to simulate the angular velocity, angular acceleration, linear velocity, and linear acceleration of the underwater vehicle.

$$\begin{bmatrix} v_b \\ w_b \end{bmatrix} = 0.1 \begin{bmatrix} \sin(0.05t) + \cos(0.01t) \\ \sin(0.01t) + \cos(0.015t) \\ \sin(0.015t) + \cos(0.01t) \\ \sin(0.03t) + \cos(0.04t) \\ \sin(0.04t) + \cos(0.03t) \\ \sin(0.05t) + \cos(0.03t) \end{bmatrix} \quad (47)$$

$$\begin{bmatrix} \dot{v}_b \\ \dot{w}_b \end{bmatrix} = 0.01 \begin{bmatrix} \cos(0.05t) - \sin(0.01t) \\ \cos(0.01t) - \sin(0.015t) \\ \cos(0.015t) - \sin(0.01t) \\ \cos(0.03t) - \sin(0.04t) \\ \cos(0.04t) - \sin(0.03t) \\ \cos(0.05t) - \sin(0.03t) \end{bmatrix} \quad (48)$$

The simulation experiments are conducted in MATLAB R2021a, using the fourth order Runge-Kutta method (ode45) to solve the differential equations, and running on a computer with an Intel(R) Core(TM) i7-12700F CPU @ 2.1 GHz and 16 GB RAM.

The parameter selection for IRTMPC-ASMC is as follows: the prediction horizon is $N = 10$, the state gain matrix $Q = \text{diag}(1, \dots, 1)_6$, the output gain matrix $R = \text{diag}(0.1, \dots, 0.1)_6$, the adaptive coefficient $q = 190$, the sliding surface gain $C = [\text{diag}(130, 130, 140), \text{diag}(1, 1, 1)]$, the discrete time $\delta_t = 0.01$, the parameter $\epsilon = 10^{-5}$. The state constraints, the output constraints and the disturbance constraints are as follows:

$$\begin{aligned} c_i &= \begin{bmatrix} \text{diag}(0.3, 0.3, 0.3, 0.1, 0.1, 0.1) \\ -\text{diag}(0.3, 0.3, 0.3, 0.1, 0.1, 0.1) \\ 0_{6 \times 6} \end{bmatrix}^T \\ d_i &= \left[0_{12 \times 3}; \text{diag}\left(\frac{1}{30}, \frac{1}{30}, \frac{1}{30}\right); -\text{diag}\left(\frac{1}{30}, \frac{1}{30}, \frac{1}{30}\right) \right]^T \\ e_i &= 2\delta_t [\text{diag}(1, \dots, 1)_6; -\text{diag}(1, \dots, 1)_6]^T \end{aligned}$$

The simulation results are presented in Fig. 4. Initially, the desired design position trajectory is plotted, representing the ideal motion path that the underwater manipulator should achieve. Subsequently, the figure showcases the actual trajectory tracking curves with three different control strategies: MPC, IRTMPC, and IRTMPC-ASMC. Notably, the tube constraint curve is also included, indicating the allowable range of fluctuation for the system state variables. Meanwhile, IRTMPC-ASMC outperforms MPC and IRTMPC, tracking the design position rapidly. Its trajectory tracking curve closely aligns with the design position curve and consistently remains within the tube constraint curve (39). This demonstrates that the control system possesses both high precision and robust stability (see Fig. 6).

The 2-norm of the error is illustrated in Fig. 5. Additionally, the figure compares the error with that of traditional MPC and IRTMPC. Clearly, the proposed IRTMPC-ASMC scheme significantly outperforms traditional MPC and IRTMPC in terms of both convergence speed and steady-state error. Detailed data on the mean square error (MSE) of each joint is presented in Table 4.

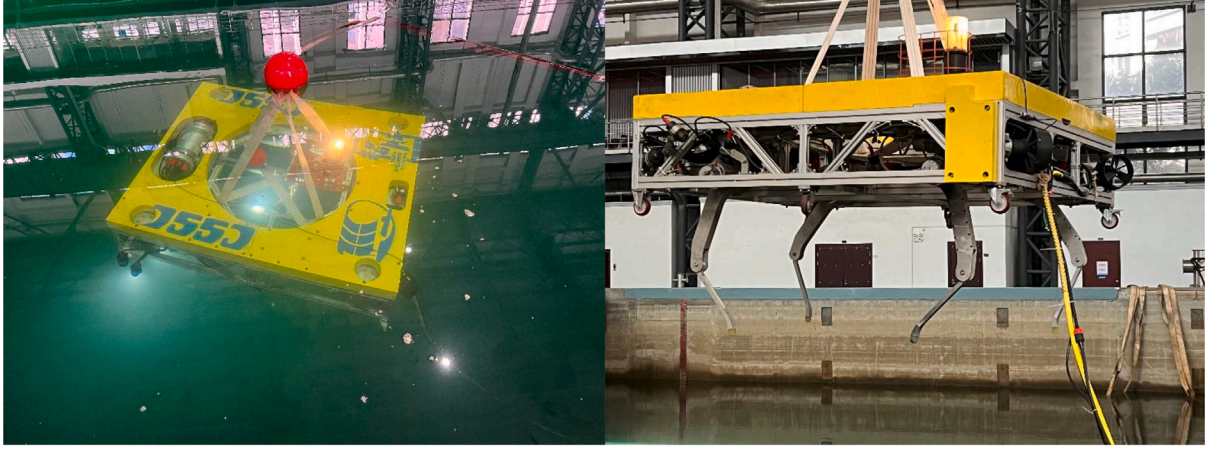


Fig. 7. ROV Equipped with Underwater Manipulators.

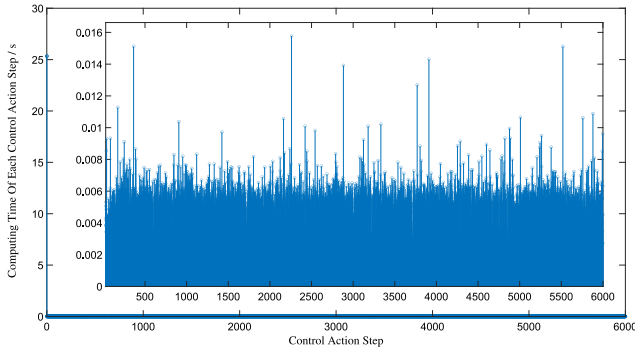


Fig. 8. Computing time of each control action step.

Table 4
Trajectory-tracking errors.

Controller	Trajectory tracking errors E_p		
	Joint 1	Joint 2	Joint 3
MPC	2.7144	2.3426	15.609
IRTMPC	2.2641	2.2226	14.152
IRTMPC-ASMC	1.8986	2.0481	13.764

$$E_p = \sum_{i=1}^n \|q_{ri} - q_i\|^2 (rad)^2.$$

The proposed IRTMPC-ASMC scheme effectively enhances the robustness of the underwater manipulator against internal and external disturbances, with large improvements in positional control accuracy for joints 1, joints 2 and joints 3. Specifically, the data shows that compared to IRTMPC, the control precision of Joint 1 increased by 16.14%, Joint 2 by 7.85%, and Joint 3 by 2.74%, with an average increase of 8.91% across all joints. When compared to MPC, the improvements are even larger, with Joint 1 increasing by 30.05%, Joint 2 by 12.57%, Joint 3 by 11.82%. This demonstrates the effectiveness of the IRTMPC-ASMC control scheme in enhancing both precision and robustness.

The joint torque variation curves are shown in Fig. 8. The torque variations are relatively smooth with no significant large-scale oscillations, and the output torques of all joints remain within the set constraint ranges.

In the specific implementation process, since the value of α needs to be sufficiently small in the calculation of the minimum positive robust set to ensure that $(A + B\Phi)^{N_s}W \subseteq \alpha W$ holds true, an appropriate value of N_s will be calculated during the algorithm initialization phase. This process requires a substantial amount of computation and will be

carried out offline. However, the N_s value will be reused in subsequent calculations to reduce the computational load. The figure below shows the code execution time for the simulation experiment, with an average execution time of less than 3 ms and a maximum execution time of less than 16 ms, meeting the real-time requirements.

6.3. Experiment

This section further conducted field control experiments on the underwater manipulator to verify the feasibility of the proposed IRTMPC-ASMC scheme in the real world. The experiments were carried out in a pool, with the studied three-degree-of-freedom manipulator mounted on a newly designed remotely operated vehicle (ROV) prototype, as shown in Fig. 7. This prototype is equipped with four three-degree-of-freedom manipulators, four horizontal thrusters, two vertical thrusters, binocular optical guidance equipment, as well as INS, a depth gauge, and DVL. These features enable the ROV to perform tasks such as stable underwater support, seabed walking, sampling in water.

This experiment aims to deploy an ROV equipped with an underwater manipulator into an experimental pool, utilizing a single manipulator to perform a predetermined trajectory tracking control task. The target trajectory set for the experiment is consistent with the excitation function (12) used in the simulation environment. The goal is to verify the control accuracy and robustness of the manipulator through actual operation. Additionally, the experiment will compare the results with those obtained using traditional Proportional-Integral-Derivative (PID) control to evaluate the performance of the proposed IRTMPC-ASMC scheme. The specific process is as follows:

- Step 1. Preparation Phase: Ensure the ROV and its underwater manipulator system are intact. Check that all connections are secure, and complete the necessary system initialization.
- Step 2. Pool Deployment: Safely lower the ROV into the pool, ensuring the manipulator is fully extended and unobstructed.
- Step 3. Trajectory Setting: Plan the manipulator's motion trajectory based on the predetermined excitation function (11) and input it into the control system of the ROV.
- Step 4. Tracking Execution: Start the manipulator and implement the IRTMPC-AMC control strategy to follow the set trajectory. Record the actual motion data and the feedback from the control system.
- Step 5. Comparison: Under the same conditions, implement the PID control strategy. Record and compare the manipulator's performance under both control methods.

The ROV has an industrial control computer (Jetson Orin NX) with an ARM Cortex-A78AEF CPU @ 2.1 GHz and 16 GB RAM in Linux operating system.

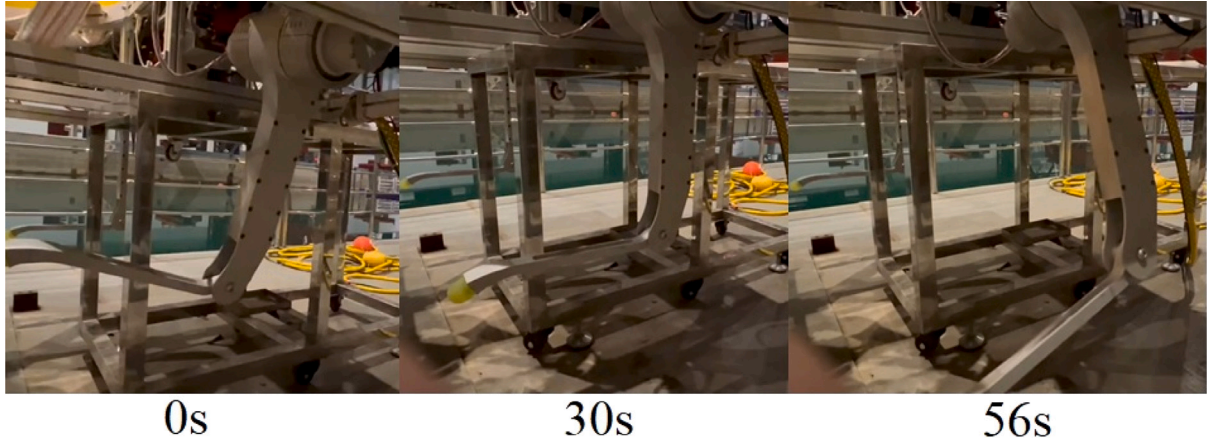


Fig. 9. Movement of the underwater manipulators.

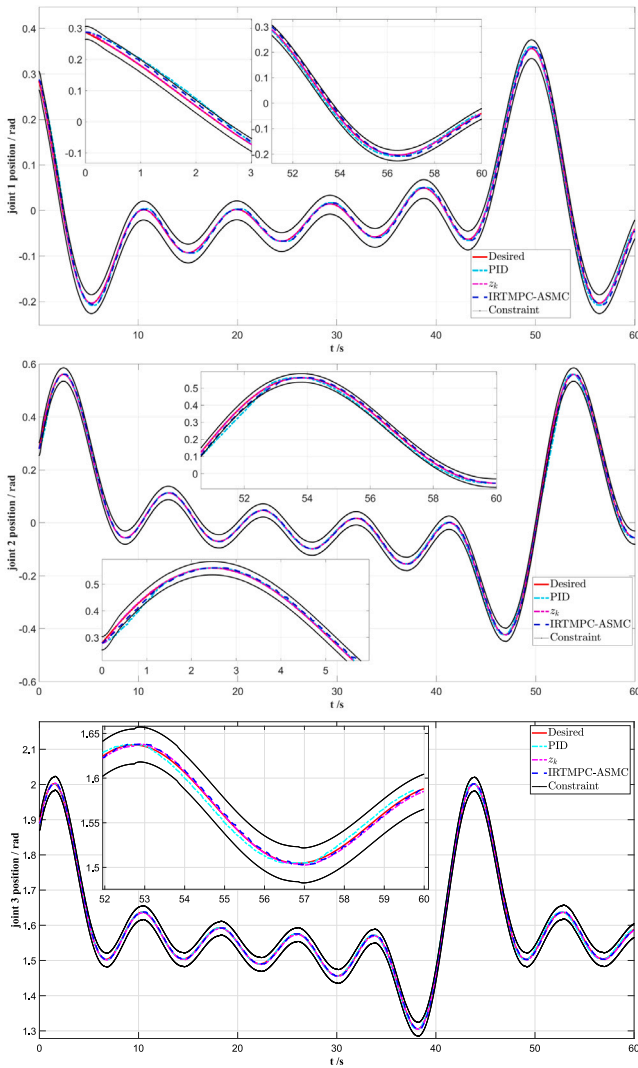


Fig. 10. The comparison of tracking by manipulator joints in real-world experiments.

The parameter selection for IRTMPC-ASMC is as follows: the prediction horizon is $N = 5$, the state gain matrix $Q = \text{diag}(1, \dots, 1)_6$, the output gain matrix $R = \text{diag}(0.1, \dots, 0.1)_6$, the adaptive coefficient

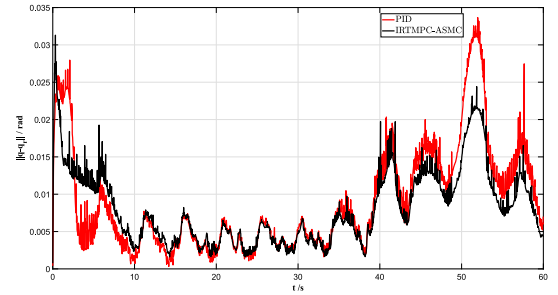


Fig. 11. The comparison of tracking error by manipulator joints in real-world experiments.

$q = 370$, the sliding surface gain $C = [\text{diag}(45, 68, 25), \text{diag}(1, 1, 1)]$, the discrete time $\delta_t = 0.02$. The state constraints, the output constraints and the disturbance constraints are as follows:

$$c_i = \begin{bmatrix} \text{diag}(0.3, 0.3, 0.3, 0.1, 0.1, 0.1) \\ -\text{diag}(0.3, 0.3, 0.3, 0.1, 0.1, 0.1) \\ 0_{6 \times 6} \end{bmatrix}^T$$

$$d_i = \left[0_{12 \times 3}; \text{diag}\left(\frac{1}{30}, \frac{1}{30}, \frac{1}{30}\right); -\text{diag}\left(\frac{1}{30}, \frac{1}{30}, \frac{1}{30}\right) \right]^T$$

$$e_i = 4\delta_t [\text{diag}(1, \dots, 1)_6; -\text{diag}(1, \dots, 1)_6]^T$$

The disturbances considered in the experiment come from two sources. Firstly, during the experiment, the ROV body is suspended in the water with positive buoyancy and is not actively controlled. As the manipulator moves, the ROV body also experiences motion, which can interfere with the manipulator's movement. The interfering force is difficult to measure and is considered as an external disturbance. Secondly, even in calm water, the manipulator still experiences complex hydrodynamic forces during movement. These forces are difficult to measure and are also treated as external disturbances.

The results of the field experiments are presented in Fig. 9, and the trajectory tracking data results are shown in Fig. 10. Initially, the desired design position trajectory is plotted. Subsequently, the figure showcases the actual trajectory tracking curves under three different control strategies: PID, IRTMPC-ASMC. Notably, the tube constraint curve is also included, indicating the allowable range of fluctuation for the system state variables. Moreover, IRTMPC-ASMC outperforms PID, tracking curve closely aligns with the design position curve and consistently remains within the tube constraint curve, showing that the control system possesses both high precision and robust stability.

The 2-norm of the error is specifically illustrated in Fig. 11. Additionally, the figure compares the error with that of PID control.

Table 5
Trajectory-tracking errors in real-world experiments.

Controller	Trajectory tracking errors E_p		
	Joint 1	Joint 2	Joint 3
PID	0.3986	0.3848	0.2365
IRTMPC-ASMC	0.3239	0.2998	0.2306

$$E_p = \sum_{i=1}^n \|q_{ri} - q_i\|^2 (\text{rad})^2.$$

Clearly, the proposed IRTMPC-ASMC scheme outperforms PID in terms of steady-state error. Especially after 40 s of motion, the superiority of the control effect becomes particularly evident. The analysis reveals that, after a longer period of motion, a complex flow field forms around the manipulator, making external disturbances more complicated and thereby increasing the tracking error. In contrast, the proposed IRTMPC-ASMC scheme's robustness is more apparent, demonstrating better control performance compared to PID control when facing complex external disturbances. Detailed data on the mean square error (MSE) of each joint is presented in Table 5.

The proposed IRTMPC-ASMC scheme effectively enhances the robustness of the underwater manipulator against internal and external disturbances, with significant improvements in positional control accuracy for all the three joints. Specifically, the control accuracy of joint 1 is increased by 18.74%, joint 2 by 22.09%, and joint 3 by 2.49%. On average, the control accuracy of all joints is improved by 14.44%, demonstrating the effectiveness of IRTMPC-ASMC scheme.

7. Conclusions

In this paper, we first performed inertia parameter identification for a newly designed underwater manipulator. More importantly, we proposed a control scheme that combines IRTMPC and ASMC for multi-input multi-output systems, using an underwater manipulator as an example. We proved that this scheme offers high tracking accuracy and robustness.

From a control perspective, we designed the scheme using discrete-time nominal MPC and discrete-time tube ASMC methods, handling state constraints, input constraints, and terminal constraints implicitly and approximately. The nominal MPC improved online convex optimization through an approximate predictive control law. The tube ASMC control law ensured constraint relationships and offered good robustness. Finally, we validated the effectiveness and robustness of the proposed IRTMPC-ASMC scheme through simulation control and field experiments.

In future work, we will focus more on the collaborative planning and control of the ROV and multiple manipulators.

CRedit authorship contribution statement

Jingming Xu: Writing – original draft, Methodology, Conceptualization. **Weidong Liu:** Writing – review & editing, Funding acquisition, Conceptualization. **Le Li:** Writing – review & editing, Funding acquisition, Conceptualization. **Guanghao Yang:** Visualization, Data curation. **Liwei Guo:** Data curation.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

References

- Calanca, A., Capisani, L.M., Ferrara, A., Magnani, L., 2011. MIMO closed loop identification of an industrial robot. *IEEE Trans. Contr. Syst. Technol.* 19 (5), 1214–1224. <http://dx.doi.org/10.1109/TCST.2010.2077294>, URL: <http://ieeexplore.ieee.org/document/5634145/>.
- Chen, S.W., Wang, T., Atanasov, N., Kumar, V., Morari, M., 2022. Large scale model predictive control with neural networks and primal active sets. *Automatica* 135, 109947. <http://dx.doi.org/10.1016/j.automatica.2021.109947>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0005109821004738>.
- Dai, Y., Gao, H., Yu, S., Ju, Z., 2022. A fast tube model predictive control scheme based on sliding mode control for underwater vehicle-manipulator system. *Ocean Eng.* 254, 111259. <http://dx.doi.org/10.1016/j.oceaneng.2022.111259>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0029801822006539>.
- Dai, Y., Yu, S., Yan, Y., 2020. An adaptive EKF-FMPC for the trajectory tracking of UVMS. *IEEE J. Ocean. Eng.* 45 (3), 699–713. <http://dx.doi.org/10.1109/JOE.2019.2899689>, URL: <https://ieeexplore.ieee.org/document/8675963/>.
- Dai, L., Yu, Y., Zhai, D.H., Huang, T., Xia, Y., 2021. Robust model predictive tracking control for robot manipulators with disturbances. *IEEE Trans. Ind. Electron.* 68 (5), 4288–4297. <http://dx.doi.org/10.1109/TIE.2020.2984986>, URL: <https://ieeexplore.ieee.org/document/9058969/>.
- Fan, J., Jin, L., Xie, Z., Li, S., Zheng, Y., 2022. Data-driven motion-force control scheme for redundant manipulators: a kinematic perspective. *IEEE Trans. Ind. Inf.* 18 (8), 5338–5347. <http://dx.doi.org/10.1109/TII.2021.3125449>, URL: <https://ieeexplore.ieee.org/document/9606588/>.
- Fleming, J., Kouvaritakis, B., Cannon, M., 2015. Robust tube MPC for linear systems with multiplicative uncertainty. *IEEE Trans. Autom. Control* 60 (4), 1087–1092. <http://dx.doi.org/10.1109/TAC.2014.2336358>, URL: <https://ieeexplore.ieee.org/document/6849468>.
- Ge, D., Wang, G., Ge, J., Xiang, B., You, Y., Feng, A., 2024. Trajectory tracking control of two-joint underwater manipulator in ocean-wave environment. *Ocean Eng.* 292, 116329. <http://dx.doi.org/10.1016/j.oceaneng.2023.116329>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0029801823027130>.
- Han, L., Tang, G., Cheng, M., Huang, H., Xie, D., 2021. Adaptive nonsingular fast terminal sliding mode tracking control for an underwater vehicle-manipulator system with extended state observer. *JMSE* 9 (5), 501. <http://dx.doi.org/10.3390/jmse9050501>, URL: <https://www.mdpi.com/2077-1312/9/5/501>.
- Han, Y., Zhang, H., Shi, Z., Liang, S., 2022. An adaptive fuzzy control model for multi-joint manipulators. *Comput. Syst. Sci. Eng.* 40 (3), 1043–1057. <http://dx.doi.org/10.32604/csse.2022.017742>, URL: <https://www.techscience.com/csse/v40n3/44562>.
- Hao, L.Y., Wang, R.Z., Shen, C., Shi, Y., 2024. Trajectory tracking control of autonomous underwater vehicles using improved tube-based model predictive control approach. *IEEE Trans. Ind. Inf.* 20 (4), 5647–5657. <http://dx.doi.org/10.1109/TII.2023.3331772>, URL: <https://ieeexplore.ieee.org/document/10352935/>.
- Hewing, L., Kabzan, J., Zeilinger, M.N., 2020. Cautious model predictive control using gaussian process regression. *IEEE Trans. Contr. Syst. Technol.* 28 (6), 2736–2743. <http://dx.doi.org/10.1109/TCST.2019.2949757>, URL: <https://ieeexplore.ieee.org/document/8909368/>.
- Janssen, N.H.J., Kools, L., Antunes, D.J., 2020. Embedded learning-based model predictive control for mobile robots using gaussian process regression. In: 2020 American Control Conference (ACC). IEEE, Denver, CO, USA, pp. 1124–1130. <http://dx.doi.org/10.23919/ACC45564.2020.9147443>, URL: <https://ieeexplore.ieee.org/document/9147443/>.
- Kang, E., Qiao, H., Gao, J., Yang, W., 2021. Neural network-based model predictive tracking control of an uncertain robotic manipulator with input constraints. *ISA Trans.* 109, 89–101. <http://dx.doi.org/10.1016/j.isatra.2020.10.009>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0019057820304043>.
- Kohler, J., Soloperto, R., Muller, M.A., Allgower, F., 2021. A Computationally efficient robust model predictive control framework for uncertain nonlinear systems. *IEEE Trans. Autom. Control* 66 (2), 794–801. <http://dx.doi.org/10.1109/TAC.2020.2982585>, URL: <https://ieeexplore.ieee.org/document/9044326/>.
- Li, Z., Li, S., 2021. A sparse optimization-based control method for manipulator with simultaneous potential energy minimization. *IEEE Trans. Circuits Syst. II* 68 (6), 2062–2066. <http://dx.doi.org/10.1109/TCSII.2020.3045427>, URL: <https://ieeexplore.ieee.org/document/9296961/>.

- Li, Y.R., Peng, C.C., Juang, J.N., 2023. An integral method for parameter identification of a nonlinear robot subject to quantization error. *Nonlinear Dynam.* 111 (24), 22419–22441. <http://dx.doi.org/10.1007/s11071-023-09027-z>, URL: <https://link.springer.com/10.1007/s11071-023-09027-z>.
- Liu, W., Xu, J., Li, L., Zhang, K., Zhang, H., 2023. Adaptive model predictive control for underwater manipulators using gaussian process regression. *JMSE* 11 (9), 1641. <http://dx.doi.org/10.3390/jmse11091641>, URL: <https://www.mdpi.com/2077-1312/11/9/1641>.
- Mao, J., Song, G., Hao, S., Zhang, M., Song, A., 2023. Development of a lightweight underwater manipulator for delicate structural repair operations. *IEEE Robot. Autom. Lett.* 8 (10), 6563–6570. <http://dx.doi.org/10.1109/LRA.2023.3309129>.
- Mayne, D., Seron, M., Raković, S., 2005. Robust model predictive control of constrained linear systems with bounded disturbances. *Automatica* 41 (2), 219–224. <http://dx.doi.org/10.1016/j.automatica.2004.08.019>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0005109804002870>.
- Nubert, J., Kohler, J., Berenz, V., Allgower, F., Trimpe, S., 2020. Safe and fast tracking on a robot manipulator: robust MPC and neural network control. *IEEE Robot. Autom. Lett.* 5 (2), 3050–3057. <http://dx.doi.org/10.1109/LRA.2020.2975727>, URL: <https://ieeexplore.ieee.org/document/9006856/>.
- Raković, S.V., 2023. The implicit rigid tube model predictive control. *Automatica* 157, 111234. <http://dx.doi.org/10.1016/j.automatica.2023.111234>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0005109823003953>.
- Ruan, D., Chen, J., Peng, X., Ai, J., 2024. Development of a full ocean depth hydraulic manipulator with heavy-duty capacity. *Front. Mar. Sci.* 11, <http://dx.doi.org/10.3389/fmars.2024.1470361>, URL: <https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2024.1470361>.
- Simetti, E., Campos, R., Di Vito, D., Quintana, J., Antonelli, G., Garcia, R., Turetta, A., 2021. Sea mining exploration with an UVMS: Experimental validation of the control and perception framework. *IEEE/ASME Trans. Mechatronics* 26 (3), 1635–1645. <http://dx.doi.org/10.1109/TMECH.2020.3025973>.
- Sousa, C.D., Cortesao, R., 2019. Inertia tensor properties in robot dynamics identification: a linear matrix inequality approach. *IEEE/ASME Trans. Mechatron.* 24 (1), 406–411. <http://dx.doi.org/10.1109/TMECH.2019.2891177>, URL: <https://ieeexplore.ieee.org/document/8603830/>.
- Xu, J., Liu, W., Li, L., 2022. An improved GA-based inverse kinematics solution algorithm for underwater manipulators. In: 2022 13th Asian Control Conference. ASCC, IEEE, Jeju, Korea, Republic of, pp. 2091–2096. <http://dx.doi.org/10.23919/ASCC56756.2022.9828177>, URL: <https://ieeexplore.ieee.org/document/9828177/>.
- Xu, P., Qin, H., Ma, J., Deng, Z., Xue, Y., 2023. Data-driven model predictive control for ships with gaussian process. *Ocean Eng.* 268, 113420. <http://dx.doi.org/10.1016/j.oceaneng.2022.113420>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0029801822027032>.
- Yang, Y., Xu, H., Yao, X., 2024. Disturbance rejection event-triggered robust model predictive control for tracking of constrained uncertain robotic manipulators. *IEEE Trans. Cybern.* 54 (6), 3540–3552. <http://dx.doi.org/10.1109/TCYB.2023.3305941>, URL: <https://ieeexplore.ieee.org/document/10242171/>.
- Yu, X., He, W., Li, H., Sun, J., 2021. Adaptive fuzzy full-state and output-feedback control for uncertain robots with output constraint. *IEEE Trans. Syst. Man Cybern. Syst.* 51 (11), 6994–7007. <http://dx.doi.org/10.1109/TSMC.2019.2963072>, URL: <https://ieeexplore.ieee.org/document/8979163/>.
- Yuguang, Z., Fan, Y., 2019. Dynamic modeling and adaptive fuzzy sliding mode control for multi-link underwater manipulators. *Ocean Eng.* 187, 106202. <http://dx.doi.org/10.1016/j.oceaneng.2019.106202>, URL: <https://www.sciencedirect.com/science/article/pii/S0029801819303877>.
- Zanon, M., Gros, S., 2021. Safe reinforcement learning using robust MPC. *IEEE Trans. Autom. Control* 66 (8), 3638–3652. <http://dx.doi.org/10.1109/TAC.2020.3024161>, URL: <https://ieeexplore.ieee.org/document/9198135/>.
- Zhang, T., Li, S., Zheng, Y., 2022. Implementable stability guaranteed lyapunov-based data-driven model predictive control with evolving gaussian process. *Ind. Eng. Chem. Res.* 61 (39), 14681–14690. <http://dx.doi.org/10.1021/acs.iecr.2c01963>, URL: <https://pubs.acs.org/doi/10.1021/acs.iecr.2c01963>.
- Zhang, A., Wang, L., Jin, J., Chen, D., Liu, R., Zhao, H., 2021. A novel piezoelectric actuated 2-DOF joint for underwater manipulator: design, simulation, and experimental investigation. *Smart Mater. Struct.* 30 (10), 105032. <http://dx.doi.org/10.1088/1361-665X/ac207c>, URL: <https://iopscience.iop.org/article/10.1088/1361-665X/ac207c>.
- Zhang, J., Zhang, F., Cheng, M., Ding, R., Xu, B., Zong, H., 2024. Parameter identification of hydraulic manipulators considering physical feasibility and control stability. *IEEE Trans. Ind. Electron.* 71 (1), 718–728. <http://dx.doi.org/10.1109/TIE.2023.3250753>, URL: <https://ieeexplore.ieee.org/document/10061487/>.
- Zheng, H., Wu, J., Wu, W., Zhang, Y., 2020. Robust dynamic positioning of autonomous surface vessels with tube-based model predictive control. *Ocean Eng.* 199, 106820. <http://dx.doi.org/10.1016/j.oceaneng.2019.106820>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0029801819309187>.
- Zhou, Y., Li, Z., Zhang, X., Li, Y., Zhu, M., 2024. A semilinearized approach for dynamic identification of manipulator based on nonlinear friction model. *IEEE Trans. Instrum. Meas.* 73, 1–20. <http://dx.doi.org/10.1109/TIM.2024.3374292>, URL: <https://ieeexplore.ieee.org/document/10462166/>.
- Zhou, S., Shen, C., Xia, Y., Chen, Z., Zhu, S., 2022. Adaptive robust control design for underwater multi-DoF hydraulic manipulator. *Ocean Eng.* 248, 110822. <http://dx.doi.org/10.1016/j.oceaneng.2022.110822>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0029801822002669>.